UV Curing Station

Category: **Power Tools & Workshop Machines** Type: **Full SOP** Priority: **High** 1 labs

Covers these lab items: UV Curing station for SLA

What it is used for

A UV curing station exposes compatible UV-sensitive resin or coating to controlled ultraviolet light to harden it. Correct exposure settings and handling help achieve complete curing while reducing UV exposure risks.

Main parts

- UV light source or LED array
- Curing chamber or enclosed exposure area
- Turntable or build-platform holder if fitted
- Timer and exposure controls
- Door, lid, or safety interlock
- Protective eyewear and resin-handling accessories

Before you start

1. Read the resin manufacturer's curing instructions and confirm the required UV wavelength.
2. Inspect the chamber, light source, door, timer, and safety interlocks for damage.
3. Ensure the station is clean and the reflective surfaces are free of residue.
4. Wear suitable gloves and UV-rated eye protection when handling uncured resin or servicing as instructed.
5. Remove uncured resin from the part using the approved washing procedure before curing.
6. Confirm the part fits inside the chamber and can receive light on the required surfaces.

How to use

1. Place the cleaned part on the designated holder or turntable.
2. Select the exposure time and other settings specified for the resin and part geometry.
3. Close the chamber door or lid fully before activating the UV source.
4. Start the curing cycle and allow the part to receive the specified UV exposure.
5. Use the turntable or reposition the part only according to the station instructions.
6. Observe the cycle from outside the chamber without looking directly at the UV source.
7. Wait for the UV source to switch off and the interlock to release before opening the chamber.
8. Remove the part using suitable gloves and check the surface and curing result.
9. Repeat exposure only when the resin instructions permit and the part requires it.
10. Switch off the station, clean any residue using approved methods, and record settings if required.

⚠ Safety

- Never look directly at an operating UV source or expose skin to its radiation.
- Do not bypass door interlocks or operate an open curing chamber unless specifically designed and shielded for it.
- Wear suitable gloves when handling uncured resin and follow its safety data sheet.
- Provide appropriate ventilation when resin products may release vapours during curing.
- Disconnect power before cleaning internal components or replacing the UV source.
- Allow hot lamps or components to cool before handling them.

Care & storage

- Keep the chamber window, reflective surfaces, and turntable clean according to the instructions.
- Inspect the UV source, wiring, seals, and interlocks regularly.
- Replace UV lamps or LED modules according to the manufacturer's service guidance.
- Store uncured resin in its recommended container away from sunlight and unauthorized access.

Common mistakes to avoid

- Using the wrong wavelength or exposure time can leave resin incompletely cured.
- Opening the chamber during exposure can expose eyes and skin to UV radiation.
- Curing a part with excess uncured resin can produce a sticky or uneven finish.
- Touching optical surfaces with unsuitable cleaners can reduce light output or damage components.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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